

Contrast Limited Adaptive Local Histogram Equalization Method for Poor Contrast Image Enhancement

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ABSTRACT Contrast limited adaptive histogram equalization (CLAHE) is a widely utilised method for image enhancement due to its speed and simplicity. However, this method faces two major limitations, namely, the requirement of manual parameter setting (i.e. clip limit and window size) and the use of a single fixed weight for enhancement, which lead to artifacts in some regions of the enhanced images. A number of approaches are used to overcome these limitations. However, each approach has its gaps, such as complexity, parameter tuning sensitivity, dependence on initial image quality and long computational time. Furthermore, most methods may perform poorly well on images with different types. This paper proposes a new modified version of CLAHE, called Contrast Limited Adaptive Local Histogram Equalization (CLALHE), to address its limitations. This method is designed to enhance image contrast locally and adaptively, without the need for user input. The CLALHE approach consists of several steps: First, multiple enhancements are applied to identify the optimal enhancement parameters. Second, these parameters are determined adaptively. Third, the original image is divided into subimages, and the optimal parameters are applied to each subimage independently, emphasizing and enhancing local features. Finally, the enhanced subimages are combined to produce the resulting image. The effects of local and adaptive concepts were systematically explored to identify the optimal parameters and validate the performance using three datasets (DIARETDB1, Pasadena Houses 2000 and Faces 1999). Qualitative assessments demonstrated the excellent performance of CLALHE and showcased enhanced images with improved contrast, better-defined details and less time consumption. Quantitative evaluation further confirmed the efficacy of CLALHE, which surpasses other methods in terms of peak signal-to-noise ratio, entropy, absolute mean brightness error, structure similarity index, contrast improvement index and root mean square error, across various image types.

INDEX TERMS Image processing, contrast enhancement, histogram equalization, CLAHE, sub-image enhancement.

I. INTRODUCTION

The swift improvements in computer vision techniques has extended the use of digital image processing systems in several domains, such as video surveillance, industrial production, remote sensing surveillance and intelligent

and other related areas [2]. However, image acquisition usually encounters uncontrollable facets (i.e. lighting conditions, camera settings, sensor noise, lens distortion, environmental conditions and angle of capture), which lead to different image weaknesses. Further, under nonoptimal lighting conditions, such as cloudy days, indoors or at night, the

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monitoring [1]. Consequently, these systems play crucial roles in daily lives, industrial sector, military applications

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amount of reflected light from the surface of an object may be insufficient [3]. As a result, low-light images may suffer from reduced image visibility and utility. Captured images in nonoptimal environments feature drawbacks, such as poor contrast, poor brightness, a narrow grey range and colour distortion, and large noise [4]. As a consequence, the implementation of image enhancement techniques is pivotal in enhancing the quality and clarity of images, which facilitates improved analysis and interpretation [5]. Image enhancement involves the improvement of the optical quality of a digital picture. This process typically encompasses various methods aimed at the increased visual appeal of an image or extraction of additional information from it [6]. Image enhancement techniques include processes, such as brightness and contrast enhancement (CE), histogram equalisation (HE), noise reduction, sharpening and colour correction. These methods can be implemented automatically through specially designed algorithms or manually through image editing software [7]. CE serves as the most rapidly growing field in digital image processing. This field is utilised in numerous applications, such as facial recognition, satellite imaging, radiographic image processing and some liquid-crystal display applications [8]. Contrast modification of an image entails the adjustment of pixel intensities to heighten the distinction between background and objects within an image [9], [10]. Image CE involves two essential categories of techniques: frequency and spatial domains [11]. Frequency domain techniques initially convert images into their frequency domain by manipulating frequency bins in the transform domain; then, image Fourier transform is computed, and the result is modified using a filter transfer function [12]. Lastly, the resulting image is obtained through inverse Fourier transform. The second method, known as the spatial domain approach, directly modifies an image's pixel values [13]. The spatial domain approach incorporates advanced techniques for image CE, such as HE, unsharp masking and grey-level grouping (GLG). According to state-of-the-art studies, which will be indicated in the following sections, most of the introduced approaches are based on the concept of HE. Hence, the introduced approaches, which adopt new visions, are highly attractive due to their role in various scientific disciplines. Furthermore, the increased demand for high-quality images necessitates the development of new and efficient enhancement techniques to improve visual perception. In addition, deep learning, especially convolutional neural networks (CNNs), has made considerable progress in computer vision. CNNs excel at feature extraction and have a high capacity due to their complex structures and many parameters. However, this increased performance relies on having more parameters and complicated designs. In addition, these models require substantial computing power and memory, which make them difficult to use in real-world applications [14], [15]. As a result, this area remains open for further research due to the various challenges involved. The primary motivation behind this study is to introduce a novel low-complexity

adaptive algorithm that can deliver

satisfactory results. In this work, we propose a new adaptive, local technique that enhances the contrast of digital images locally, without requiring user expertise. The sections of this paper are organised as follows: Section II reviews related methods based on CHE technique. Section III discusses the conventional CLAHE methodology. Section IV discusses variants of CLAHE. Section V introduces the proposed methodology. Section VI describes the proposed approach's contribution. Section VII covers the data sample and measurement metrics. Section VIII presents the results and discussion. Section IX discusses the computational complexity. Finally, Section X concludes.

II. LITERATURE REVIEW

Images with a sufficient contrast typically reveal more details than those with a low contrast [16]. In regards to the enhanced contrast of grayscale and colour images, the approaches used fall under two primary categories: direct and indirect [17]. Direct enhancement methods straightforwardly modify image intensities through the application of a defined contrast value, whereas indirect methods adjust intensities across a certain dynamic range without using a specific contrast value. Hence, traditional enhancement techniques often fail to improve a broad spectrum of low-contrast images. In response, a multitude of research have been prompted to develop new solutions for various image processing challenges. In the spatial domain, state-of-the-art methods are used to enhance image contrast. These methods encompass HE, unsharp masking and GLG. Conventional HE (CHE) is designed to even out and improve image intensity distribution for the effective improvement of image quality. This method achieves such a goal by modifying an image's cumulative density function (CDF), which helps in balancing the grayscale intensities across it and leads to an enhanced contrast [18], [19]. Grey levels with less probabilistic densities are combined to a singular grey level, and the gap between adjacent grey levels expands if they possess a substantial probabilistic density. This phase generates a uniform intensity distribution in the improved image. Essentially, HE improves the overall contrast and simplifies image enhancement [20]. This concept has been extensively applied in recent approaches because of its effectiveness, low computational time and simplicity [21]. However, the images obtained through this method display distortion and washed-out characteristics, which improve input-image mean values [22]. The final images exhibit a nonuniform brightness distribution in the background and minor areas, particularly in cases with numerous grey levels with high frequencies. Furthermore, this method fails to retain the information present in input images, which leads to the loss of fine details at low grey levels and contrast distortion at specific grey levels [23]. Based on literature review, this study categorises the CHE method into six classes: global HE (GHE), local HE (LHE), bi-subimaging HE (BSHE), multi-subimaging HE (MSHE), weighted HE (WHE) and exposure region HE (ERHE). GHE improves

the entire contrast of an image; this straightforward, efficient enhancement method [24] enhances the dynamic range of grey levels and redistributes the bin density to improve image contrast. Furthermore, an image's CDF serves as the basis for CE in GHE. This technique is easy to use, efficient and time efficient; hence, it is widely employed in tracking services and real-time scenarios [25]. However, its main problem is the washed-out images resulting from the shifting effect of the mean value of the original image [19], [26]. Moreover, the concentration of high-frequency components at certain grey levels, which encompasses small patches and the background and typically occurs at other low-frequency levels, suggests that GHE cannot retain the original image details. This limitation at low levels leads to conflicting constraints and the loss of some fine details [23]. In GHE, the dynamic clipped HE (DCLHE) is used to enhance low-contrast images [27]. This approach involves the elimination of intensity levels with zero density from the image histogram prior to the utilisation of a minimal histogram value to reduce nonzero density grey levels, which consequently improves contrast. DCLHE can preserve an image's entropy and produce a uniform grey level distribution [27]. However, DCLHE cannot maintain image brightness, and hence, it can only be used within a limited range of histogram grey levels. LHE overcomes the shortcomings found in GHE. It notably improves the contrast within local greyscale segments of an image to ensure detail preservation. This category encompasses several techniques aimed at fulfilling these goals, including adaptive HE (AHE) [24], CLAHE [28], partially over-loaded sub-image HE (POSHE) [29], cascaded multistep binomial filtering HE (CMBFHE) [30], adaptive image enhancement method using contrast limitation based on multiple layers block overlapped HE (BOHE) [31], iterated adaptive entropy clip limit (CL) HE (IAECHE) [32] and adaptive entropy index HE (AEIHE) [33]. The AHE algorithm processes individual image pixels, which results in a uniform-brightness image that effectively addresses the nonuniform illumination encountered in the GHE method [24]. However, this method increases noise, requires a long computation time and results in an unnatural-looking image [29]. CLAHE, POSHE and CMBFHE also occasionally produce poor appearance and noise effects [34], [35]. Meanwhile, POSHE possesses a slightly improved computation time. POSHE effectively reduces noise in images and improves the clarity of local details [31]. These outcomes are achieved through multilayer enhancement, which involves the rapid application of BOHE to various-sized windows derived from the original image. These windows effectively reduce noise through a contrast limit method. However, several drawbacks have been associated with this technique, including long computation time, poor capability to conserve the texture of the input image and an inability to enhance the resulting images' illumination. Moreover, IAECHE determines the CL by partitioning an image into several contextual blocks and then performing calculations using the peaks from each

IAECHE determines the optimal CL by utilising an iterative and adaptive method in its calculations. Subsequently, a final image is generated through amalgamation of the processed blocks. However, IAECHE faces two limitations, i.e. prolonged computation time and inadequate capability to converge at the optimal CL. In AEIHE, images are initially divided into three subgroups to enhance local features [33]. Then, the contextual region and CL for individual subimages are determined adaptively while considering their structure and informational quality. Subsequently, the high information quality and structure of the resulting subimage are preserved through the introduction of a new parameter termed the entropy index. The limitations of IAECHE include its incapability to emphasise local details and its extensive computation duration. The third subclass known brightness standard deviation HE, addresses the limitation of GHE by generating two subhistograms using the original image histogram. This approach aims to improve illumination preservation. Various iterations of BSHE, such as brightness-preserving bi-HE (BBHE) [36], dual subimage HE (DSIHE) [37], minimal mean brightness error bi-HE (MMBEBHE) [38], Otsu-based BBHE bi-HE (OBBHE) [39], entropy-based bi-HE (EBBHE) [39], bi-HE-based three plateau limits (BHE3PL) [40] and range-limited bi-HE (RLBHE), have been developed to further enhance its functionality [41]. BBHE [36] and DSIHE [37] can produce two subhistograms utilising the median and mean of the input histogram, respectively, to maintain image brightness. Meanwhile, MMBEBHE [38] achieves the same goal by segmenting an image's histogram and identifying a threshold value that minimises the 'minimum brightness error'. Similar to GHE, both approaches are susceptible to excessive emphasis of high-frequency components, particularly when a subhistogram contains multiple prevalent levels. This tendency often results in the introduction of unmanageable noise [27], [42]. MMBEBHE also requires a prolonged processing time [38]. Otsu and entropy are applied in OBBHE [39] and EBBHE [39] using the histogram threshold of an image. Concerning image brightness maintenance, these methods generate better images than BBHE, DSIHE and MMBEBHE. Nevertheless, these methods suffer from inefficient image CE [43]. BHE3PL [40] can produce images of excellent and superior quality. RLBHE also uses Otsu [41]; it divides an input-image histogram to two subhistograms and preserves brightness by separating the foreground from the background; however, the process washes out image details [44]. The fourth subclass, MSHE, is further subclassified as follows: recursive mean separate HE (RMSHE) [45], recursive subimage HE (RSIHE) [46], median-mean subimage-based clipped HE (MMSICHE) [47] and adaptive thresholding-based sub-HE (ATSHE) [48]. The RMSHE and RSIHE algorithms operate in recursive mode and maintain image brightness through multiple decompositions, with the median and mean values serving as thresholds. However, the manual setting of recursive level based solely on visual assessment results in increased noise and processing

time due to the recurring decomposition process. Considerable improvements in the image only at notably high recursive levels. Meanwhile, MMSICHE assigns a recursive level a value of 2 and calculates the mean to generate four subhistograms using the original histogram. Subsequently, MMSICHE computes the median to trim each subhistogram. The sub-histograms are then applied with CHE. ATSHE identifies the subhistogram number at each stage through the initial computation of the peak signal-to-noise ratio (PSNR) and determines the thresholds using standard deviations and means. Subsequently, ATSHE clips and enhances image contrast using the histogram's median. However, ATSHE shows an association with certain drawbacks: manual counting of the number of iterations, which results in prolonged computation time and insignificant improvements; blurring of certain image regions, which leads to the loss of intricate details; the need for a comprehensive evaluation of its efficiency. Histogram compression, washout and other important overenhancing problems have been resolved using dynamic HE [49] and brightness-preserving dynamic HE (BPDHE). Dynamic HE involves histogram segmentation of an input image to multiple subhistograms based on local minima values; this process is continued until the subhistograms no longer show a discernible dominant part. Although this method effectively preserves all image details, the resulting image may display overbrightness [19]. BPDHE applies a Gaussian filter and a local maximum to partition an image histogram. Although it effectively reduces the extreme brightness caused by dynamic HE, BPDHE yields subhistograms with varying ranges. Although most of these subhistograms exhibit wide grey level ranges, a small subset demonstrates constrained ranges. As a result, nonuniform image enhancement may occur across all subhistogram ranges [13]. The quadrants for dynamic HE (QDHE) serves as an alternative method that prevents noise amplification and excessive contrast enhancement in images while simultaneously preserving image details [50]. The histogram is subdivided into multiple subhistograms, and the clipping value is adjusted through QDHE by calculating the mean of each subhistogram. Subsequently, a new dynamic range is created using the dynamic HE features derived from the resultant subhistograms. AIVHE [51] redistributes the original probability density function (PDF) to prevent considerable alterations to the PDF and provides better CE. Parameter specification in AIVHE requires human intervention, which is a major drawback. By contrast, an entropy-based dynamic sub HE (EDSHE) was proposed to address the shortcomings of RMSHE [52]. This entropy-based divergence similarity metric (EDSHE) calculates the entropy for all recursive layers and compares the results with the initial value. However, in comparison with the compound entropy CHE and RMSHE metrics, EDSHE generates images with numerous artifacts and requires a long processing period. The fifth subclass, known as WHE, controls the increase in the pre dominant grey level, which entails the modification of the

image histogram. These techniques involve the adjustment or clipping of high-frequency histogram levels to lessen their prevalence while conserving image features. A comparison of the final image produced by WHE with those generated by the three previously mentioned contrast harmonic mean (CHE) subclasses revealed that the modified histograms diminish the importance of dominant grey levels. Within the framework of WHE, various CE techniques include the modified HE (MHE) [23], weighted threshold HE (WTHE) [53], recursively separated and WHE (RSWHE) [54], range-limited WHE (RLWHE) [55], edge preservation LHE (EPLHE) [56], a logarithmic law-based histogram modification scheme for naturalness image CE [57], a bi-histogram modification method for images with nonuniform illumination and low contrast [58], image enhancement via a subimage HE based on mean and variance (MVSHE) [42], weighted average multisegment HE (WAMSHE) [59] and high-speed quantile based HE for brightness preservation and CE (HSQHE) [60]. MHE improves the final image while preventing artifacts in BBHE, DSIHE and RMSHE. This technique involves CHE to improve contrast subsequent to the modification of the original image's histogram. However, MHE is limited by its subjectivity in determining the optimal parameter value and potential loss of information due to the reduction of high-frequency histogram levels. Meanwhile, WTHE applies CHE to the modified PDF derived from alteration of the original image's PDF values. This method allows for customization to control the overenhancement of images. However, although WTHE has a fast processing time, it often fails to maintain the original image brightness, which can lead to the creation of unwanted artifacts. RSWHE comprises two components: RSWHE-M and RSWHE-D. The former employs a mean value computation to partition a histogram in a manner comparable to that used for RMSHE, and the latter uses the median to divide a histogram. Subsequently, a normalised power-law distribution function is used to adjust these subhistograms. Then, each subhistogram is processed using traditional HE. Although RSWHE-D and RSWHE-M notably enhance contrast and preserve intensity, the resulting image may exhibit undesirable defects caused by the loss of image information [27], [51], [55]. RLWHE is applied to manage CE and improve brightness. This approach operates swiftly and applies to real-time scenarios. The original image histogram is split into two subhistograms to lessen intraclass variance. Each subhistogram's PDF is adjusted and weighted, and its brightness is controlled using CHE. Adaptive gamma correction boosts the information density of an image and lessens unwanted defects. A homomorphic filter is utilised in the final enhancing stage to lessen the effect of artifacts. EPLHE theorises that if a subimage is within a zero-textured area, POSHE fails to preserve the image object edges. As a result, the area has a low information richness, and the CDF values exhibit inadequate changes, which lead to over- or under-enhancement [56]. The EPLHE technique partitions the input image to nonoverlapping subimages of

equal sizes. Subsequently, the Sobel operator is applied to determine the gradient percentage within subimages and effectively identify object edges. One limitation of EPLHE is its reliance on the input image's scale during subimaging. This method often retains extreme brightness in dark areas, which results in the failure to maintain detailed image information. Furthermore, it cannot provide sufficient quantitative measurement criteria [61]. The LMHE-proposed approach involves a logarithmic law-based histogram that modifies the natural image contrast method in two steps to preserve the intriguing aspects of an image [57]. In the initial phase, an additional-based approach is used to address histogram pits, and the subsequent phase utilises a logarithmic-law based strategy to mitigate the problem of histogram spikes. After the two stages, CHE is employed to improve the over all contrast of an image before fine tuning the same image locally via DCT. Nonetheless, this approach fails to conserve image brightness, which potentially leads to darkened areas within an image [58]. This method includes three functional steps. In the initial phase, noise amplification is prevented through the application of a Gaussian low-pass filter, which also facilitates the extraction of illumination from the image. Subsequently, median is applied in threshold calculation to segregate the final image to dark and bright areas, and the histogram of each region is adjusted accordingly. The darker section is shifted to the segmentation threshold to increase brightness and prevent inconsequential subsequent image appearances. Meanwhile, the brighter section is adjusted to the optimal weight between the median and mean. The next step involves the partition of the histogram of each region into segments for modification and redistribution based on change function. Although this method reduces noise intensification and preserves image details, the final image may exhibit fuzzy areas and artifacts regardless of the enhanced image entropy. The MVSHE separates the histogram of an original image into four sections using three adaptive thresholds based on variance and mean. Then, the CHE method is employed to adjust and equalise each segment. For decreased intensity saturation and interference, the method merges the enhanced and normalised segments. Furthermore, MVSHE yields an image while maintaining its mean brightness and eliminating any artifacts. Nonetheless, this method requires an extended processing time and fails to preserve or highlight information in white areas or fourth histogram section, which hampers its capability to unveil hidden details. Alternatively, in WAMSHE, the histogram of an image is divided into several subhistograms before equalising each one [59]. This method minimises unwanted noise and artifacts in image processing while preserving the overall brightness levels. In HSQHE [60], the quantile (q) parameter is utilised to segment an image's histogram into an equal number of subhistograms. Each quantile is then individually adjusted, equalised and normalised. Subsequently, the entire histogram is equalised and normalised to produce the output

image. The undesired artifacts and distortion in the subhistogram may be caused by low intensities in the quantile, which requires

a thorough comprehension and subjective determination of the ideal value of q , which relies heavily on user experience. The sixth CHE subclass, ERHE, involves the segmentation of a histogram into distinct regions. ERHE generates output images that are optimally enhanced without being excessively or inadequately modified, which improves the illumination of images across various regions. Within the framework of ERHE, various CE techniques, such as exposure-based subimage HE (ESIHE) [62], mean-based bi-HE plateau limit (BHEPL) [63], median-based BHEPL [63], adaptive bi-HE (ABHE) algorithm [63] and exposure-region-based multi HE (ERMHE) [64], are used to improve image brightness, mostly of images with a nonuniform brightness. The image histogram is divided into two segments using an exposure threshold based on the ESIHE technique, followed by the clipping of both subhistograms with the mean grey level. Subsequently, CHE equalises these subhistograms. Although this method effectively enhances contrast and preserves image details, it is unsuitable for images with multiple exposure zones. BHEPL and ABHE operate with two exposure areas (i.e. underexposure and overexposure) to ensure compatibility with ESIHE. Regardless, these methods do not work in the typical exposure region. ERMHE addresses over-, normal and under-exposure in the multiexposure region to provide an image with improved brightness preservation and less noise. Notably, information loss may occur in specific regions due to a high-density dominance across low frequencies of grey levels, and one region may be smaller than the other. Table 1 summarises the CHE-based CE techniques.

Based on Table 1, three issues must be considered:

- i. Various categories of CE techniques have been developed to specifically target certain issues associated with digital images. Techniques, such as GHE, are aimed at redistributing intensity levels to achieve a uniform contrast while overlooking the improvement of unhighlighted or hidden image features. Conversely, the LHE category operates on the opposite concept of GHE. Each category addresses specific image enhancement requirements rather than tackle all common issues simultaneously.
- ii. Most contemporary techniques require manual parameter tuning, which needs a high computation time and relies on the expertise and experience of the user. Nonoptimal parameter tuning may lead to an unsatisfactory final image, which results in either an under- or overenhanced input image.
- iii. Numerous methods, particularly those under the LHE category, have been developed to enhance local image details. However, these approaches are often constrained by their tendency to produce unnatural results.

Overall, the current CHE methods suffer from several draw

backs. Notably, washout and detail loss prevail in images generated by methods under the GHE subclass. Meanwhile, methods under Consequently, these methods are unsuitable for critical

Enhancement **TABLE 1.** Summary of CHE-based techniques for CE.

undesired artifacts and require lengthy computation times. In addition, the methods included in MSHE and BSHE subclasses encounter challenges related to determining the optimum subimaging level and exhibit prolonged computation periods. In high-frequency dominant regions, approaches belonging to the WHE subclass suffer from preserving image features and information. The above-described issues offer bases for contributing a novel LHE technique to enhance image contrast. The suggested approach is aimed at preserving image structure by emphasising local features and improving its information richness locally and adaptively. Thus, the proposed approach overcomes the second obstacle through the automatic and adaptive adjustment of all parameters. This condition results in the removal of subjective effects and offers an objective means of optimising contrast distribution over the resultant image to ensure uniformity. Furthermore, the proposed method improves local image characteristics without creating undesired visual distortions or increasing noise levels.

III. CLAHE

An image’s contrast can be improved without the occurrence of undesirable artifacts, such as the ‘halo effect’ associated with AHE, by employing CLAHE [65]. In cases involving image regions with a narrow histogram, classical AHE can potentially overly heighten contrast, which results in pronounced noise and artifacts in specific regions [66]. This phenomenon, known as the ‘halo effect’, transpires as a distinct grid-like pattern surrounding edges [67]. CLAHE addresses this issue by constraining the CE within each image tile. Histogram restriction is accomplished by capping at

the LHE subclass compromise image brightness, introduce

tile. These histograms are utilised to enhance image contrast. As a result, the resulting images from CLAHE exhibit superior entropy values, which enhance the visual quality of images [73]. However, despite these advantages, the CLAHE approach faces some shortcomings:

- The process involves enhancing the intensities of foreground and background image areas at an equal weight, which results in an output image with highly contrasted regions. This noise amplification in uniform areas imparts an unnatural structure to the output image [74].
- The input image is segmented into small regions called tiles or tile size (TS) to enhance minor variations in grey levels and prevent loss of image details. However, this segmentation process prolongs the creation of an enhanced image [69].

Setting the enhancement parameters TS and CL in CLAHE is subjective and largely based on the user's prior knowledge [75]. When determining the CL, a small CL number may enable a minor enhancement of image contrast, whereas opting for a large CL can cause pixel distribution across the grey level histogram to flatten and amplify noise.

The CL settings must be adjusted based on specific requirements of individual images. In addition, the TS substantially influences the natural quality of the resulting images. A low TS aids in averting noise amplification in the final image and preserving less local image features; a high TS value amplifies noise and retains more local image details, which

a defined value referred to as the 'clip limit'. This measure averts excessive amplification of noise and artifacts, which ensures that the CE in areas with a narrow histogram remains within appropriate bounds [68]. Figure 1 illustrates the concept map of conventional CLAHE, and Figure 2 shows CLAHE's pseudo code. The CLAHE technique improves images by initially dividing them into small areas called tiles.

For each tile, CLAHE calculates a histogram representing pixel distribution. It then clips these histograms at a predetermined 'clip limit' to reduce excessive contrast. After clipping, the histograms are equalised, and the most frequent intensity values are spread. Image reconstruction is then achieved using the modified values from these processes. CL was selected based on user skills and knowledge to fine tune CE [69]. This method is valuable in medical imaging field, such as in X-ray and computed tomography (CT) scans [70], where it unveiled subtle details that are otherwise hard for the human eye to detect. Beyond medical applications, CLAHE is also employed in satellite imagery [71], which showcases its

potentially results in output images with undesirable characteristics [24]. Furthermore, the application of the default settings for CLAHE parameters (TS and CL) may result in an artificial appearance and excessive contrast in the enhanced images. Therefore, the optimal parameter values must be carefully selected when utilising CLAHE to ensure the production of high-quality enhanced images.

IV. VARIANTS OF CLAHE

According to a bibliometric study based on the Scopus database, 873 papers on CLAHE have been published in various journals from 2014 to 2024. 'CLAHE' was the utilised search query, the subject area was limited to 'Engineering'

FIGURE 1. Concept map of CLAHE framework. **FIGURE 2.** Pseudo code of the

CLAHE algorithm.

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versatility in improving image clarity across various domains. A detailed explanation about CLAHE can be found in another work [72].

Compared with the AHE approach, the CLAHE approach exhibits adaptiveness and capability to reduce noise amplification through histogram clipping and redistribution using a predefined value prior to CDF calculation [32]. Unlike the AHE approach, CLAHE partitions the input image into multitriles before the calculation of distinct histograms for each

and 'Computer Science', the keywords were limited to 'CLAHE', 'Contrast Limited Adaptive Histogram Equalisation (CLAHE)', 'Contrast Limited Adaptive Histogram

Equalization' and 'Contrast-limited Adaptive Histogram Equalization', and the language was limited to English. The number of published papers considerably increased

FIGURE 3. Publication interest in CLAHE techniques over the last 10 years based on scopus database.

from 2014 to 2024 (Figure 3). Several researchers contributed substantially efforts to the massive progress in research areas regarding CLAHE from different approaches over the last 10 years (i.e. 2015 to 2024). A group of researchers contributed hybrid-based methods, and others modified the CLAHE framework by adjusting its structure or its parameter settings through an adaptive procedure to enhance the performance of CLAHE and make it effective for different image types.

A medical fundus image enhancement technique [76] was presented by integrating features of filtering and CLAHE (denoised and CLAHE). The proposed model aims to enhance contrast and denoising at the R, G and B components of the fundus image. This previous study showed that improving the contrast and lowering noise in retinal fundus images was more effective than other methods. The results were of high quality, appeared natural and required less processing time. The method achieved notable improvements in PSNR, structural similarity index measure (SSIM), coefficient of correlation (CoC) and edge preservation index (EPI) compared with the state-of-the-art method, which indicate that it preserves edges better in enhanced images. The proposed method offers several advantages. The integration of filtering techniques with CLAHE remarkably enhances the contrast and clarity of retinal fundus images. The method effectively removes various types of noise, including Gaussian (white) and salt-and-pepper (impulse) noise. The use of CLAHE with filtering techniques helps in the preservation of important details in the fundus images. However, the proposed method has some limitations. The method involves multiple steps and applies various filters, which can increase computational complexity and processing time. The performance of the method highly depends on the choice of parameters, such as the type of filter used and settings for CLAHE. The initial quality of input images contributes to the effectiveness of the enhancement. The approach may

the proposed method enhances the contrast and local details of images, which result in their easier interpretation and analysis. The CLAHE-PLT method demonstrated excellent performance in enhancing contrast and reducing noise in images. Moreover, the results show notable improvements in visual quality, PSNR, information entropy (IE), luminance contrast (LC) and mean squared error (MSE) compared with conventional methods. The CLAHE-PLT method has several advantages and limitations. The method considerably improves the contrast and brightness of ultrasonic well logging images, which results in a better visual quality without visible artifacts. Moreover, the method achieves a need modifications to suit different types of images.

Authors in [77] came up with a new means to improve ultrasonic well-logging images by combining CLAHE with power-law transformation. They called it CLAHE-PLT. Thus,

high accuracy in image quality assessment metrics, such as PSNR, IE and LC, outperforming other state-of-the-art techniques. However, the CLAHE-PLT method has notable limitations; its performance can be sensitive to the choice of parameters, such as the gamma value and CL, and thus requires careful tuning for optimal results. Furthermore, while effective for low-contrast images, the method may perform poorly on images with various types of degradation or artifacts.

ACE-HM is an adaptive CE technique based on a histogram modification framework that improves traditional CLAHE [78]. It dynamically modifies the histogram of each image block based on local content, which ensures balanced enhancement across various regions. The results showed notable improvements in contrast enhancement while preserving natural image quality, preventing overenhancement in homogeneous areas and maintaining important image details. The suggested method achieves higher entropy and EME values than CLAHE, HE and other methods. Thus, image details are better preserved, and the method makes image viewing easier. The advantages of this proposed method include adaptive histogram modification based on local image content, which results in better-balanced CE. In addition, this method can preserve details. Bilinear interpolation also reduces blocking artifacts, which eliminates the problems caused by block-wise histogram processing and makes the transitions between image regions smoother. Despite its advantages, this method has several limitations. It requires additional processing for tasks, such as adaptive histogram modification, conditional probability calculations and mapping, which make it more computationally intensive than simpler techniques, such as HE and CLAHE. Furthermore, the method's performance is affected by parameters, including the threshold (T) for conditional histogram modification and λ for histogram balancing. Incorrect tuning of these parameters can

negatively affect the quality of the enhancement. In addition, although the method aims to preserve image details, it may inadvertently amplify noise in very noisy images with notable variations in pixel intensities.

Authors in [79] introduced a method for enhancing under

water images by combining CLAHE with homomorphic filtering (HF). This method, called CLAHE-HF, uses spatial and frequency domain improvements to improve underwater images. It attains such a goal by fixing problems, including

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uneven lighting, low contrast and noise. By integrating these techniques, the method effectively enhances underwater images. The results of the CLAHE-HF method revealed crucial improvements in visual quality compared with traditional enhancement techniques. Specifically, it achieved a higher PSNR of 29.3435, a lower MSE of 69.8760 and a higher entropy (7.4826). This method effectively enhances image contrast, clarity and detail while reducing noise, which make it a valuable tool for underwater image processing. The CLAHE-HF method has several advantages and limitations. The method substantially improves the contrast and clarity of underwater images, which results in better visual quality without visible artifacts. The enhanced images maintain local details and texture, which avoids the over-enhancement and unrealistic contrast often observed with other techniques. Meanwhile, the method involves multiple steps and calculations, which may increase the complexity of its implementation compared with simpler techniques. The performance of the method may be sensitive to the choice of parameters, such as the CL for CLAHE and filter parameters for HF, and requires careful tuning for optimal results. The method may require more computational resources compared with simpler enhancement techniques, which can limit real-time applications on devices with a limited processing power. The initial quality of underwater images plays a crucial role in the overall performance. Extremely degraded images may still pose challenges for enhancement.

A novel method for enhancing poor-contrast images called IAECHC was proposed [32]. The method aims to produce an output image with the best entropy value to enhance poor-contrast images. It effectively enhances image details, preserves brightness and reduces noise, which make it a valuable tool for the enhancement of poor-contrast images. The IAECHC results demonstrate superior performance in enhancing image contrast and quality. Moreover, it showed significant improvements in visual quality, PSNR, entropy and structural similarity index (SSI) compared with other methods. The IAECHC method offers several advantages, with the resultant images displaying a better contrast, clearer details and more natural appearance compared with those of other methods. The IAECHC method adaptively calculates parameters, such as the CL and clip factor, which ensure that the enhancement process is tailored to each subimage. The method fine tunes the enhancement process through multiple iterations, which leads to superior results. Despite all these advantages, the IAECHC suffers from major limitations. The IAECHC method involves multiple steps and calculations, which make it more complex to implement compared with simpler techniques. The iterative process and adaptive parameter

calculation add to its complexity. Although the method produces high-quality results, it requires more computational time due to the iterative enhancement process. The performance of the method may be sensitive to the choice of parameters, such as the window size (w) and C value. Careful tuning is required to achieve optimal results. The initial quality of the input image contributes to the overall performance

of the method. Extremely degraded images may still pose challenges for enhancement, even with the IAECHC method. Authors in [80] developed the agent-based image contrast enhancement (ABICE) algorithm. This algorithm treats each pixel in an image as an independent agent that adheres to specific rules aimed at improving the image's contrast. The algorithm's iterative process gradually boosts contrast until it reaches the desired level. The resulting images exhibit substantially improved contrast and features and appear more natural and visually appealing compared with those processed by alternative methods. Furthermore, the ABICE algorithm demonstrates lower execution times relative to other techniques, which highlights its computational efficiency. The ABICE algorithm presents several advantages and limitations. ABICE tremendously improves the contrast and clarity of images, which make critical details and features more visible. The algorithm achieves excellent scores on performance metrics, such as SSIM, EPI, entropy, Relative Enhancement.

Contrast (REC) and Range Redistribution (RR), which means that it retains more details and improves contrast. The ABICE algorithm demonstrates lower execution time compared with other methods, which make it suitable for real-time applications. ABICE has some limitations because its algorithm requires several steps and interactions between agents. These limitations can make it more complicated to implement than simpler methods. The performance of the algorithm may be sensitive to the choice of parameters, such as the intensity modification factors in the behavioural rules. Careful tuning is required to achieve optimal results. The agent-based approach may require more memory to store and process the interactions between agents, which can be a limitation for devices with limited memory resources.

A low-light image enhancement method based on the CLAHE approach called adaptive contrast limit CLAHE (ACL-CLAHE) was proposed to enhance the quality of unmanned aerial vehicle (UAV) images obtained under low-light conditions to enable robust object detection [81]. The proposed method combines CLAHE with colour space conversion and gamma correction to enhance low-light images. The resultant images show considerably improved contrast and visibility of structural details. The images appeared more natural and had reduced noise compared with those processed with traditional methods. The ACL-CLAHE outperformed conventional methods and

achieved higher natural image quality evaluator scores, which indicate a better image quality. The method demonstrated efficient implementation, with an average runtime of 4.05 ms per image. The ACL-CLAHE offers several advantages; it substantially improves the contrast and brightness of low-light UAV images, which results in an

improved visual quality without visible artifacts. The method boosts object detection accuracy. By using an optimised contrast limit, the method effectively reduces noise amplification, which is a common issue in traditional CE techniques. Moreover, the method is computationally efficient. Despite these

advantages, ACL-CLAHE has numerous limitations. The method involves several process steps, which may increase the complexity of its implementation compared with simpler techniques. Furthermore, the performance of the method depends on the choice of parameters, such as the contrast limit (ϕ) and gamma value (γ). Careful tuning is required to achieve optimal results. Although the method is effective for low-light images, it may perform poorly on images with various types of degradation or artifacts. The method specifically aims to improve UAV images of power transmission systems. Furthermore, although the method is efficient for individual images, processing large datasets or high-resolution images may still require considerable amount of time and computational resources.

Another proposed method [82] integrates independent component analysis (ICA) with CLAHE (ICA & CLAHE) to enhance image quality and feature recognition while preserving details. The ICA & CLAHE method yields significant results, with the enhanced images showing better contrast, clarity and detail preservation compared with those processed by traditional methods. Furthermore, the proposed method achieved higher PSNR values, which indicates better image quality with reduced noise. The proposed method has several advantages. The method notably improves the visual quality of images by enhancing contrast and preserving details. The synergy of ICA and CLAHE improves feature recognition in images. ICA enhances the separation of signal components, and CLAHE improves contrast, which result in clearer and more distinct features. However, this method holds several limitations. It involves multiple steps and calculations, including the application of ICA and CLAHE, which make its implementation more complex compared with simpler techniques. The method may also require more computational resources due to the integration of ICA and CLAHE, which can be a limitation for real-time applications or devices with limited processing power. The performance of the method may be sensitive to the choice of parameters, such as the separation matrix and fusion coefficients in ICA and the CL in CLAHE. Careful tuning is required to achieve optimal results.

An advanced method [83] was introduced to enhance the contrast of remote sensing images (RSIs). The proposed technique, which is inspired by CLAHE, aims to optimise CE by incorporating a dynamic and automated approach. The results demonstrated notable improvements in CE for RSIs compared with the traditional CLAHE technique. The method enhances image details more effectively, particularly in bright areas, while maintaining a natural appearance. The proposed method presents several advantages, as it

dynamically calculates a unique CL for each contextual region based on the presence of outliers. The proposed technique remarkably reduces the number of excess pixels that need to be redistributed, which helps preserve more information from the original image. Moreover, the proposed method outperforms CLAHE across various quantitative metrics, with higher scores achieved in contrast, EME and entropy. Despite these

advantages, this method has various limitations. The introduction of additional blocks for outlier detection and dynamic CL calculation may increase the computational demand compared with the standard CLAHE. The effectiveness of the proposed method relies on the accurate detection of outliers within each contextual region. Poor outlier detection may lead to suboptimal CL calculations, which potentially affects the overall enhancement quality.

The authors in [84] proposed a method to enhance retinal fundus images for better diagnosis of glaucoma using two CLAHE techniques (i.e. enhanced CLAHE (En-CLAHE) and automated CLAHE (Auto-CLAHE)). These methods aim to improve contrast, reduce noise and maintain the natural appearance of images by dynamically adjusting the CL and automating the tile selection process. The results confirmed the effectiveness of the En-CLAHE and Auto-CLAHE methods in assisting the diagnosis of glaucoma and other retinal diseases, with these methods achieving higher PSNR and entropy values compared with traditional CLAHE. En-CLAHE and Auto-CLAHE have several advantages and limitations. En-CLAHE dynamically adjusts the CL based on the local histogram of each subimage, which effectively reduces noise while enhancing contrast. Auto-CLAHE automates the tile selection and CL determination processes, which make it more user friendly and adaptable to different images. The enhanced images maintain a natural look and avoid overenhancement and unrealistic contrast, which is crucial for accurate medical diagnosis. Despite these advantages, En-CLAHE and Auto-CLAHE have major limitations. Both methods involve multiple steps and calculations, which may increase the implementation complexity compared with simpler techniques. The performance of En-CLAHE and Auto-CLAHE may be sensitive to the choice of parameters, such as the CL and the number of tiles, and thus require careful tuning for optimal results. Although Auto-CLAHE automates the enhancement process, it may require 1–2 min to process an image on a typical machine, which can be a limitation for high-volume clinical settings. Furthermore, the initial quality of the retinal fundus images contributes to the overall performance.

Table 2 summarises various methods for image enhancement using different versions of CLAHE. Each study integrated CLAHE with additional techniques, such as filters, gamma correction or fuzzy logic to improve image quality,

visibility and detail preservation in various types of images, including CT scans, X-rays, MRI brain images, retinal fundus images and ultrasonic well logging images. The results demonstrated notable improvements in visual quality, contrast, noise reduction and preservation of natural appearance and local details. However, the methods suffer from several limitations, such as high complexity, parameter sensitivity and dependency on the initial image quality. Some methods may perform poorly on different image types, and their computational requirements may limit real-time applications. Thus, this study proposed a new CLAHE-based CE method to address such problems.

TABLE 2. Summary of variants of CLAHE.

ADAPTIVE CONCEPT METHODOLOGY

The proposed method, contrast limited adaptive local histogram equalization (CLALHE), is a new approach for the local enhancement of poor-contrast images based on an adaptive concept. Given the varying areas of contrast in each image type, CLALHE hypothesises that their enhancement weights must be suitably tuned for each area. This hypothesis was proposed because the foreground area contains essential objects, followed by the region of interest (ROI), which holds worthy image information. Meanwhile, the background contains less worthy and crucial information, such as the sky,

the sea, and the wall and the cleared or black background areas in the medical field. The proposed method not only aims to improve the richness of information and details within an image but also preserve its overall structure.

The proposed CLALHE method varies from the conventional CLAHE in terms of the following criteria: (i) an adaptively and locally based concept for parameter computation based on input image information; (ii) selection process for optimal parameters; (iii) subdivision process of the input image to various subimages and separate processing of each.

This section discusses the theories related to the proposed CLALHE. In accordance with the criteria, the algorithm

V. PROPOSED CLAHE BASED ON LOCAL AND

FIGURE 4. Block diagram of the proposed CLALHE.

improves the overall image contrast and enhances local details to produce a visually pleasing image and preserve image structure. Figure 4 illustrates a graphical representation of the proposed method, with more information provided in subsequent subsections. The proposed CLALHE method comprises two parts, namely, (i) Part 1: determining optimum parameters and (ii) Part 2: obtaining a resultant image. As the proposed CLALHE concept is developed to be a local and adaptive enhancement technique. The first part of the proposed CLALHE method focuses on the computation of the image histogram to extract its peaks and valleys, which are further utilised to calculate other parameters. Furthermore, this part aims to identify the optimum parameters for Part 2 criteria. The second part of the proposed CLALHE method is an essential part, where the input image is subdivided into subimages and hence improved individually, beginning with the first subimage and ending with the last one based on the optimum parameters provided previously in Part 1. The proposed CLALHE differs from the conventional CLAHE in terms of the following:

(a) In Part 1, three parameters are introduced: (i) context block dimension (CBD), (ii) N -parameter and (iii) ideal CL (I_{CL}). CBD refers to the optimum dimensions of the context regions for CLAHE (i.e. width and height). The N -parameter is introduced as a new parameter, and it plays a crucial role in calculating a specific value used in the I_{CL} parameter's formula, which ensures that the I_{CL} remains within acceptable limits. An overly high I_{CL} can result

FIGURE 5. Flowchart of the proposed CLALHE.

in the overenhancement of the resultant image, which must be avoided. Lastly, the I_{CL} parameter is introduced as an equation to calculate the CL used in the proposed method.

(b) In Part 2, two parameters are proposed: (i) optimum number of subimages (ONS) and (ii) subimage dimensions (SID). The ONS parameter is designed to obtain an even number, which is used to identify the best number of subimages to be used in subdividing the input image equally.

Moreover, the ONS is determined based on the number of valleys in the input image's histogram. The SID is designed to calculate the dimensions of subimages based on the ONS , and it is used to subdivide the input image to subimages. These proposed parameters in Parts 1 and 2 guarantee the local and adaptive enhancement of input images to produce the best enhanced final image without washed-out or unwanted artifact details. Figure 5 depicts the operation flowchart of the proposed CLALHE method.

A. PART 1: DETERMINING OPTIMUM PARAMETERS This section discusses Part 1 of the proposed CLALHE technique in detail. Part 1 includes six crucial steps: (i) reading of the input image; (ii) acquisition of the image histogram; (iii) identification and quantification of the number of histogram peaks and valleys; (iv) computation of parameters

Enhancement **FIGURE 6**. Input image of the (a) original and (b) greyscale images.

FIGURE 7. Histogram distribution of the greyscale image in Figure 6 (b).

(N -parameter, CBD , and I_{CL}); (v) application of CLAHE; (vi) sorting of the optimum enhancement parameters. Once an input image is provided, it will be converted into a greyscale scheme (Figure 6). By converting the RGB image to greyscale, we can apply the CLALHE method efficiently (in terms of processing time) because we only need to apply the technique to a single channel, which represents the image's intensity (greyscale). The greyscale image retains the spatial information of the RGB image but without the colour information, which allows us to focus on enhancing the image contrast without dealing with multiple colour channels.

Figure 6 demonstrates the input image, and Figure 7 illustrates the histogram distribution of the input image. The next step is the identification of the number of peaks and valleys of the histogram. To identify a histogram peak, we

compared

the frequency (i.e. the pixel number) of each grey level in the histogram to those of its neighbouring grey levels on the right and left sides. If the frequency of a particular grey level was higher than that of its neighbours, then the pixel was identified as a peak. The number of peaks was then determined and denoted as N_{Peaks} . Furthermore, for the identification of histogram valley, the frequency (i.e. the number of pixels) of each grey level in the image histogram was compared with those of its neighbouring grey levels on the right and left sides. If the frequency of a particular grey level was lower than that of its neighbours, then the pixel was considered a valley. The number of valleys is then quantified and denoted as $N_{valleys}$.

To illustrate the identification process of histogram peaks and valleys, we selected a range of grey levels (i.e. 0 to

FIGURE 8. Identified peaks and valleys for the selected sample (i.e. 0–20 grey levels). Note: The red triangle refers to the valley, and the green triangle indicates the peak.

7 and 15 for peaks and 2, 14 and 16 for valleys.

20) from the histogram distribution of the grey image in Figure 7. In accordance with the definition of identifying histogram peaks and valleys, we illustrated the detected peaks and valleys (Figure 8). After the identification and quantification of the peaks and valleys numbers, their values were used as bases for the calculations of the newly introduced parameters (i.e. N -parameter, CBD and L_{CL}) in the next step. The N -parameter was calculated adaptively using the following equation:

This illustration identified 2 peaks and 3 valleys. Table 3 tabulates the identified peaks and valleys, with grey levels of

TABLE 3. Identified peaks and valleys in Figure 8. (0–20 grey levels).

$$P_{Valley[i]}$$

$$N_{Valleys}(1)$$

$$N - parameter =$$

where $Valley[i]$ represents the frequency (i.e. pixel number) of each identified grey level as a valley in the image histogram, and $N_{Valleys}$ refers to the number of identified valleys in the image histogram. Adaptive calculation of the CBD parameter is achieved using the following:

$$CBD = \frac{Width}{Height} = \frac{\lceil \ln(N_{Peaks}) \rceil}{\lceil \ln(N_{Peaks}) \rceil} \quad (2)$$

where in the expression $\lceil \ln(N_{Peaks}) \rceil$, $\lceil \cdot \rceil$ indicates the ceiling function, and it is used to round the natural logarithm of the variable N_{Peaks} to the nearest whole number. If the natural logarithm of N_{Peaks} , denoted as $\ln(N_{Peaks})$, results in a noninteger, then it is adjusted

upward to the closest integer. This method ensures that the output values are integers with a nonlinear growth. Let us consider a case where the number of N_{Peaks} is 52. Using Eq. (2), CBD will have the following

$$Height = \lceil \ln(52) \rceil = \lceil 3.951 \rceil = 4$$

$$Width = \lceil \ln(52) \rceil = \lceil 3.951 \rceil = 4$$

dimensions:

$$CIQI = (PSNR + Entropy)$$

$$AMBE^{(4)}$$

The value of 3.951 was rounded up to 4, and the *CBD* dimensions was set to 4×4 .

The *I_{CL}* parameter was calculated adaptively using the following equation:

$$I_{CL} = (Peaks[i] + N - parameter)$$

$$Max(Valley)^{(3)}$$

where *Peaks[i]* represents the frequency (i.e. pixel number) corresponding to each grey level identified as a peak in the image histogram. The term *Max(Valley [i])* indicates the

- By summing the values of PSNR and entropy, the equation recognises the importance of noise reduction and information content in the enhanced image and ensures that it is clear and detailed.
- The division by AMBE introduces a penalty for excessive brightness distortion, which ensures that the enhancement process does not excessively alter an image's natural appearance. This balance is crucial for applications requiring visual fidelity and information preservation.

highest valley value within the group of identified valleys. Once all required parameters for the application of the current procedure become available, the input image is subjected to the enhancement process multiple times. The frequency of enhancement process depends on the number of identified peaks (i.e. N_{Peaks}) in the image histogram. Therefore, this part tests each value (i.e. frequency) of the identified peaks group to specify the peak that generates the optimum enhanced image. To ensure that Part 1 will produce an optimum enhanced image, we proposed a new fitness function for this objective called the comprehensive image quality index (CIQI):

tion, and richness of information is image's natural appearance. The CIQI enhanced and highlighted. A low AMBE provides the following advantages: value ensures that the enhancement 1. Balancing of Image Quality Metrics: process does not excessively alter the

$$ONS \approx$$

- PSNR is widely used in the assessment of the fidelity of image processing techniques, particularly in terms of noise reduction. High PSNR values indicate the proximity of the enhanced image to the original, unprocessed image, which contributes to the maintenance of image integrity.
- Entropy estimates the quantity of information in an image, with high values denoting rich detail and desir

According to Eq. (4), the CIQI function is contrived to highlight and enhance the enrichment of subimage information and local details without structural distortion. Entropy highlights and enhances the richness of local details and information, whereas the PSNR highlights noise reduction in the image. In addition, AMBE is used to highlight brightness consistency. The proposed CLALHE requires high PSNR and entropy and the lowest AMBE value to obtain the highest or optimal CIQI. High PSNR and entropy ensure noise reduc

In conclusion, the proposed CIQI equation offers a balanced and comprehensive metric for performance estimation of the proposed CLALHE method (Part 1). The equation helps in evaluating the proposed CLALHE in an informed and nuanced manner through the combination of noise reduction, detail preservation and brightness consistency into a single metric.

Lastly, the specific parameters of the optimum enhanced image (i.e. *CBD* and *I_{CL}*) are stored for further processing in Part 2.

B. PART 2: OBTAINING A RESULTANT IMAGE Part 2 represents the final section of the proposed CLALHE method, which involves five crucial steps: (i) calculation of *ONS*, (ii) calculation of *SID*, (iii) subdivision of the input image to multiple subimages, (iv) utilisation of the optimum enhancement parameters obtained previously in Part 1 for the enhancement of each subimage; (v) combination of the enhanced subimages to produce the final enhanced image.

The first step involves the calculation of the *ONS* based on the $N_{Valleys}$ represented in the following equation:

$$\left(\log_2(N_{Valleys}) \text{ if } ONS \text{ is an even number} \right)$$

$$\log_2(N_{Valleys}) + 1 \text{ if } ONS \text{ is an odd number (5)}$$

able contrast. This condition ensures the attainment of an informative and visually appealing enhanced image.

- AMBE refers to a measure of differences in the brightness between input and enhanced images. A low AMBE indicates that enhancement preserved the natural brightness of an image, which is crucial for maintaining visual consistency.

2. Complementary Evaluation:

From Eq. (5), the *ONS* is considered even and odd integer numbers. Therefore, if the numbers of his togram valleys ($N_{Valleys}$) are 50 and 100, respectively, the

$\log_2(N_{Valleys})$ produces values of 5.65 and 6.65, respectively. Thus, the *ONS* values were set to 6 (i.e. $\lceil 5.65 \rceil = 6$) and 8 (i.e., $\lceil 6.65 \rceil + 1 = 8$). The *ONS* obtained from Eq. (5) was then used to adaptively subdivide the input image into subimages. In this study, *ONS* was set to be an even number

only. Thus, when Eq. (5) produced an odd number, 1 was added to convert the answer into an even number. Figures 9 and 10 demonstrate two examples of subdivision process for $NS = 2$ and $NS = 4$, respectively. Let us consider an image with dimensions $M \times N$ (i.e. M and N denote the numbers of columns and rows, respectively) subdivided into *ONS* subimages. After implementation of the procedure, the dimensions of each subimage (*SID*) were calculated as

FIGURE 9. Image division with $ONS = 2$ (a) right and (b) left subimages.

FIGURE 10. Image division with $ONS = 4$ (a) first, (b) second, (c) third and (d) fourth subimages.

follows:

$$SID(Height, Width) = \begin{cases} \lceil M \rceil, & \text{if } NS = 2 \\ N/NS, & \text{if } NS > 2 \end{cases} \times 2 \quad (6)$$

image.

For the image shown in Figure 9, each subimage has the following dimensions when using Eq. (5) (with original dimensions of 592×844):

$$Height = 592 \quad Width = \frac{896}{2} = 448$$

Thus, the *SID* for each subimage is (592, 448). On the other

FIGURE 11. Pseudocode of the proposed CLALHE method.

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construct the resultant
if NS = 2
    enhanced, they are then combined to
    This approach ensures that the most suitable enhancement parameters are applied to each sub-image. The resulting image is a composite of enhanced subimages

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hand, for the image shown in Figure 10, each subimage has the following dimensions:

$Height = \frac{592}{2} = 296$ $Width = \frac{896}{4 \times 2} = 448$ Thus, the *SIS* for each subimage is (296, 448). The next step includes the separate enhancement of each subimage using the optimum parameters (i.e. *CBD* and *I_CL*) identified in Part

1. Once all the subimages have been enhanced and offers an improved overall image quality. Figure 11 illustrates the pseudocode of the proposed CLALHE.

VI. CONTRIBUTION OF THE PROPOSED CLALHE METHOD The proposed CLALHE method theorises that dividing an image into subimages will help expand the intensity distribution range while processing the local features of each subimage. Unlike the conventional CLAHE

method, the proposed CLALHE introduces a locally and adaptively based approach, sorting the optimal enhancement parameters to ensure that each subimage is efficiently enhanced before being combined to create the final image. Additionally, the approach adaptively computes these parameters, as outlined in Part 1: Determining Optimum Parameters. The proposed CLALHE overcomes and mitigates several disadvantages

TABLE 4. Comparison of the proposed CLALHE versus the conventional CLAHE approach.

inherent in the conventional CLAHE method, as summarized in Table 4.

The novelty and advantages of CLALHE are as follows:

- (i) Multiple Enhancement Concept: We introduce a multi-enhancement process that enhances the input image multiple times. This approach helps identify the optimal enhancement parameters by evaluating each enhanced image.
- (ii) Adaptive Parameter Tuning: We propose a new fitness function called the Comprehensive Image Quality Index (CIQI), which integrates PSNR, entropy, and AMBE. CIQI facilitates the evaluation and identification of the optimal parameters (i.e., CBD and I_CL) for CLALHE without requiring user expertise.
- (iii) Image Subdivision: We introduce the concept of dividing the input image before applying the algorithm, allowing for local processing and better preservation of local details. This contrasts with CLAHE, which applies a uniform enhancement across the entire image.
- (iv) Local Enhancement for Each Subimage: The identified optimal parameters are applied to enhance each subimage independently. This process ensures the preservation and enhancement of local details while minimizing unwanted artifacts. The enhanced subimages are then integrated to form the final image.
- (v) Real-Time Applicability: CLALHE is particularly use

ful for various image-processing applications, especially in real-time scenarios, due to its low computational time.

VII. DATA SAMPLE AND MEASUREMENT METRICS The proposed method was assessed using 821 sample images drawn from three distinct datasets. The first dataset, Faces 1999, comprises 450 samples with dimensions of 896×592 [85]. The second dataset, Pasadena-Houses 2000 [86], consists of 241 samples sized at 1760×1168 , and the third dataset, DIARETDB1 [87], contains 130 samples measuring 1500×1100 . These datasets were obtained from the Image Processing Place, California Institute of Technology database, and Standard Diabetic Retinopathy Database. The datasets were selected due to their diverse array of image types, which encompass various levels of information richness. Some images featured extensive uniform backgrounds, such as those depicting the sky or sea, and thus contain small yet pertinent details. Meanwhile, the others showcased intricate sections alongside uniform areas. This diversity ensured that the evaluation encompassed various levels of image quality and information richness.

The proposed CLALHE was compared with seven recently developed methods: CLAHE [28], POSHE [29], BHM [58], IAECH [32], AEIHE [33], Auto-CLAHE [83] and ACL CLAHE [81]. These methods were selected based on their similarities in approach to the proposed CLALHE:

- (i) All those methods were derived and modified from CHE.
- (ii) These methods employ a similar concept of dividing an image into subimages after the local enhancement each subimage contrast.
- (iii) These methods can preserve or enhance image details. The most recent HE-based CE methods include BHM, IAECH, AEIHE, Auto-CLAHE and ACL-CLAHE, which were published in 2018, 2020, 2021, 2023, and 2024 respectively. The study encompassed qualitative and quantitative evaluations to appraise the efficacy of the proposed technique. The evaluation criteria included contrast, information retention, brightness, image structure and naturalism. Qualitative estimation involves scrutinising the visual quality of the final images, with a primary focus on CE, naturalness and over-enhancement. The original image's naturalness and details must be upheld throughout the improvement process, and a favourable overall contrast must be ensured without introducing undesirable artifacts.

The image's noise level must be maintained below a specified threshold. Visual assessment proves to be a valuable criterion for the evaluation of the proposed approach. In addition, quantitative assessments using benchmarks, such as PSNR, entropy, AMBE, SSI, CII and root mean square error (RMSE), were adopted to compare the proposed approach with other CE (CHE)-based methods.

Discrete entropy (DE) serves as a metric for image quality; it estimates the amount of information present in an image [42], [88]. A high entropy image contains abundant valuable information; therefore, a positive correlation exists between image richness and entropy. Contrarily, a high noise level may occasionally influence and result in a high DE. Hence, qualitative analysis is necessary for these cases to assist in addressing high DE value identification due to noise or information richness. DE is expressed as follows:

$$DE = - \sum_{l=0}^{L-1} p(l) \log_2(p(l)) \quad (7)$$

where $p(l)$ represents the histograms' PDF, and l denotes the grey levels of images. If the probability distribution values of all images have the same intensities, then the image entropy will be optimal, that is, $p(0) = p(1) = p(2) \dots = p(L-1) = 1/L$ [88].

The AMBE quality factor is used to measure the capacity area.

of a technique to preserve the image's mean brightness [19], [62]. The AMBE value is determined through measurement of the differences between the mean

brightness of the original and output images using the following equations:

$$AMBE = I(w, h)_{mean} - O(w, h)_{mean} \quad (8)$$

$$\mu_y = \frac{1}{T} \sum_{i=1}^T Y_i \quad (14)$$

where w and h refer to the count of pixels in the images' rows and columns, respectively; $I(w, h)_{mean}$ indicates the mean of

$$\sum_{i=1}^T X_i$$

$$X_i - \bar{X} \quad (15)$$

$$\sigma_x^2 = \frac{1}{T-1} \sum_{i=1}^T (X_i - \bar{X})^2$$

input images, and $O(w, h)_{mean}$ represents the mean of output

images. A small difference between the original and resultant images implies that the final image preserves the brightness

$$\sum_{i=1}^T Y_i$$

$$Y_i - \bar{Y} \quad (16)$$

of the original [42].

PSNR is another key metric for the assessment of the improvement between the original and resultant images; this metric calculates the MSE to determine the degree of degradation [42], [89]. PSNR can be computed using Eq. (10):

$$SSI(w, h) = (2\mu_x\mu_y + c1)(2\sigma_{xy} + c2)$$

$$MSE = \frac{1}{W \times H} \sum_{h=1}^H \sum_{w=1}^W (I(w, h) - O(w, h))^2 \quad (9)$$

$$X^W_{w=1}$$

$$PSNR = 10 \log_{10} \left(\frac{(Max(I_i))^2}{MSE} \right) \quad (10)$$

$$(Max(I_i))^2 MSE$$

$$h)^2 \quad (9)$$

small constants.

RMSE serves as a

metric for the disparity

relationship with image

quality. A low RMSE

indicates an output image

the RMSE exhibits an inverse with great detail and reduced

assessment between the original and enhanced images. In this context,

distortion compared with the original image [75].

where $(Max(I_i))^2$, W , H , $I_i(w, h)$ and $O(w, h)$ refer to

value is minimal. Eqs. (9) and (10) show the inverse relationship of MSE and PSNR. As a result, a low MSE value decreases noise amplification, improves image quality and increases PSNR [40]. The CII assessment factor serves as a method for the evaluation of the efficacy of techniques through comparison of the contrast levels of the processed and input images [90], [91]. This comparison is expressed through the following equations:

$$CII = \frac{C_{Improved}}{C_{Original}} \quad (11)$$

where $C_{Improved}$ and $C_{Original}$ denote the average of contrast for a ROI in the processed and original images, respectively. Eq. (12) is used to compute the images' contrast:

$$C = \frac{f+b}{f+b} \quad (12)$$

where f represents the mean grey level (MGL) of a specific aspect of an image, and b denotes the MGL of the surrounding

The SSI is an evaluation metric used to assess the likeness between two images. Its value ranges between [0, 1], which signifies a complete dissimilarity when approaching 0 and an exact match at 1. A high SSI value typically denotes a superior image quality [46], [92]. Eq. (17) is utilised to calculate SSI.

$$(\mu_x^2 + \mu_y^2 + c1)(\sigma_x^2 + \sigma_y^2 + c2) \quad (17)$$

where μ_x and μ_y represent the averages of images x and y , respectively, and σ_x^2 and σ_y^2 are variances. $c1$ and $c2$ represent

The RMSE is calculated as follows:

$$RMSE = \sqrt{\frac{1}{W \times H} \sum_{i=1}^W \sum_{j=1}^H (I(i, j) - O(i, j))^2} \quad (18)$$

where W and H represent the image rows and columns, respectively. Meanwhile, $I(i, j)$, and $O(i, j)$ indicate the rows and columns of the original and resulting images, respectively.

MSE was used to quantify the mean intensity difference between the input and output images. The enhanced image exhibited less deterioration than the input image if the MSE

FIGURE 12. Resultant image of human face produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECHE, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

FIGURE 13. Resultant image for the magnified yellow boxed region of the human-face image produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECHÉ, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

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FIGURE 14. Resultant image for magnified blue boxed region of the human-face image produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECHÉ, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

FIGURE 15. Resultant image for magnified purple boxed region of the human-face image produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECHÉ, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

FIGURE 16. Resultant image for magnified orange boxed region of the human-face image produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECH, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE, and (i) the proposed CLALHE.

specifications: Intel(R) Core (TM) i7-10750H CPU @ 2.60 GHz 2.59 GHz, 16 GB RAM, SSD hard disk

The proposed CLALHE and all the testing methods were simulated on a personal computer with the following

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FIGURE 17. Resultant image for magnified green boxed region of the human-face image produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECH, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

and Spyder (Python 3.8) as the integrated development environment.

VIII. RESULTS AND DISCUSSION

To validate the proposed CLALHE capabilities, we conducted qualitative and quantitative assessments. The selected state-of-the-art techniques for performance evaluation included CLAHE, POSHE, BHM, IAECH and

AEIHE. The optimum parameters recommended by the respective authors of these methods were utilised in the implementation process. Qualitative analysis provided valuable insights and a comprehensive familiarity with the performance of the proposed CLALHE. We selected three datasets (i.e. faces 1999, Pasadena-Houses 2000 and DIARETDB1) to perform this analysis and obtained one image from each dataset (Figures 12, 19 and 23, respectively) to discuss the qualitative performance in detail. The selected samples were labelled as human-face, House

01 and DIARETDB1-001. Complementing the qualitative analysis, the quantitative analysis outcomes for the three images were tabulated in Tables 5–7. The values highlighted in bold and underlined in the tables denote the best and second-best findings, respectively. Table 8 contains the average of quantitative analysis findings on all images for all examined datasets, and Table 9 includes detailed findings from the analysis of average computational times.

The first sample (i.e. human face) illustrated in Figure 12(a) shows a close shot of a human face captured in an indoor location with nonuniform illumination. Poor lighting

FIGURE 18. Resultant image for magnified red boxed region of the human-face image produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECHE, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

compromised the image quality, which resulted in obscured or inadequate background details. Figure 12 depicts the resultant image produced by the proposed CLALHE and other selected testing techniques. For a more detailed evaluation, six coloured boxes areas were indicated (i.e. orange, green, yellow, blue, purple and red) in Figures 13–18.

The CLAHE and Auto-CLAHE methods produce a final image with an over-enhanced contrast of the original image and unwanted artifacts in the background region. This finding can be noticed in the wall and door region behind the woman in Figures 12(b) and 12(g), respectively. Moreover, the woman's skin colour was different compared with that in the original image. In Figures 13(b) and 13(g), the yellow boxed region (i.e. woman's hair) is over-brightened, which is undesirable. The blue boxed region (i.e. irises) in Figures 14(b) and 14(g), and the purple boxed region (i.e. nose) in Figures 15(b) and 15(g) revealed excessive enhancement. Moreover, the red boxed region displayed unwanted noise in the door area (Figures 18(b) and 18(g)). All these observations indicate that CLAHE and Auto-CLAHE are not the best options for the enhancement of such an image type.

POSHE and ACL-CLAHE produced a resultant image with excessive brightness, which led to a loss of the woman's facial details (Figures 12(c) and 12(h)). The resultant images of the yellow, blue and purple regions indicate that these methods are unqualified for enhancing the considered image

FIGURE 19. Resultant image of House 01 produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECH, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

type because they failed to preserve the woman's hair details and the natural appearance of the chair on the left side (Figures 13(c) and 13(h), 14(c) and 14(h); 15(c) and 15(h)).

The BHM method also produced a resultant image with excess brightness. The white areas became whiter, and the woman's hair region lacked information richness and appeared washed out (Figure 12(d)). The yellow, blue and purple areas in Figures 13(d), 14(d) and 15(d) also respectively display excessive brightness and diminished details. Thus, the BHM technique produced the poorest image compared with the other methods.

The IAECH and AEIHE techniques produced enhanced images with noticeable artifacts, especially in the face area (Figures 12(e) and 12(f)). The iris colour was overenhanced and appeared close to black (Figures 14(e) and 14(f)), and

the nose region was excessively darker versus that in the input image (Figures 15(e) and 15(f)). Moreover, the door area was enhanced heterogeneously, as illustrated in the red-boxed region in Figures 18(e) and 18(f), where some areas appeared brighter and others dark. Overall, both techniques failed to effectively enhance the ROI and local details, which resulted in slight enhancements with noticeable artifacts.

The proposed CLALHE method showed a superior capability in the production of a resultant image with reduced noises or artifacts in the heterogeneous regions (i.e. the woman's shadow on the door and the wall behind her; Figure 12(i)). Moreover, the CLALHE method effectively enhanced and preserved the fine details of the resultant image, specifically the irises, nose and woman's hair, the cabinet behind the woman and the door. Furthermore, the boxed regions (i.e. yellow, blue, purple and red) illustrate

TABLE 5. Quantitative analysis of human-face image.

TABLE 6. Quantitative analysis for House 01.

enhancement and detail preservation (Figures 13(i), 14(i), 15(i) and 18(i)).

Based on the qualitative analysis detailed above, the proposed CLALHE method unveiled its superior capability in producing high-quality images with reduced artifacts, which makes it effective in detail preservation compared with other test techniques. These findings are corroborated by those of quantitative analysis. Table 5 tabulates the quantitative analysis findings, where the CLALHE method exhibited the highest PSNR and AMBE, which demonstrates its high capability to enhance and preserve image details. Moreover, the CLALHE method yielded the second-best SSI, which is close to 1 (0.991). This finding signifies the method’s capability to preserve the resultant image structure ver sus the input image. In conclusion, the combination of qualitative and quantitative analyses based on the ‘human

FIGURE 20. Resultant image for the magnified yellow boxed region of House 01 produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECH, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

face image’ verifies CLALHE’s proficiency in producing pleasing and natural looking images and exceeding in CE and local detail preservation compared with other examination techniques.

Given that the original image is captured outdoors, the second sample (i.e. House 01; Figure 19(a)), featured large nonuniform regions, including background (i.e. the sky), foreground (i.e. grass and trees) and large detailed areas (i.e. house, garden walls and car). However, the image suffered from low brightness, which resulted in the obscurity of impor tant details. Thus, this image was selected as a representative sample to evaluate the capabilities of all state-of-the-art tech nologies. For further evaluation, the analysis focused on three

FIGURE 21. Resultant image for the magnified red boxed region of House 01 produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECHÉ, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

magnified regains, denoted by coloured boxes (i.e. yellow, red and blue) in Figures 20–22.

FIGURE 22. Resultant image for the magnified blue boxed region of House 01 produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECHÉ, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

The CLAHE and ACL-CLAHE methods produced an overly enhanced resultant images; these methods over-brightened the entire region of House 01 (Figures 19(b) and 19(h)). The resultant images appeared unpleasant and unnatural to the human eye, and issues of over brightening occurred in the three boxed regions (i.e. trees, grass and the house) (Figures 20(b) and 20(h), 21(b) and 21(h) and 22(b) and 22(h), respectively). These observations prove

FIGURE 23. Resultant image of DIARETDB1-001 produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECH, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

that CLAHE and ACL-CLAHE are not the best options for enhancing the considered image type.

The POSHE method produced a resultant image with an unpleasant and unnatural appearance and suffered from distortions and artifacts (the sky part in Figure 19(c)). The magnified regions (i.e. yellow, red and blue) indicate the poor qualification of the POSHE method for the enhancement of the considered image type because it failed to conserve image information (Figures 20(c), 21(c) and

22(c)).

The BHM method produced a poorly enhanced resultant image (Figure 19(d)) with undesirable artifacts, which led to a low brightness in specific details (i.e. the grass at the forefront, left side of the house and the entrance pathway), as shown in the yellow, red, and blue boxed regions in Figures 20(d), 21(d) and 22(d), respectively.

The IAECH, AEIHE, and Auto-CLAHE methods produced slightly improved resultant images, specifically the large, detailed area and background (i.e. sky and house) in Figures 19(e), 19(f) and 19(h), respectively. However,

FIGURE 24. Resultant image for the magnified red boxed region of I. M. Mohammed, N. A. M. Isa: CLALHE Method for Poor Contrast Image Enhancement

FIGURE 25. Resultant image for the magnified orange boxed region of DIARETDB1-001 produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECH, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

DIARETDB1-001 produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECH, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

these methods failed to preserve crucial foreground region details, such as grass, trees and entrance, which led to over-enhancement of the images. Overall, these techniques struggled to effectively highlight local details, which led to slight enhancements combined with observable artifacts,

as indicated by the boxed regions evident (i.e. the yellow, red and blue) in Figures 20(e), 20(f) and 20(h), 21(e), 21(f)

and 21(h), and 22(e), 22(f), and 22(h).

The proposed CLALHE method proved its superior capability in the production of resultant images with reduced noise or artifacts while preserving the structure of the

overall image (Figure 19(i)). Moreover, it effectively enhanced and preserved local details in crucial regions, as indicated by the coloured boxed regions (i.e. yellow, red and blue) in

Enhancement **TABLE 7**. Quantitative analysis of a DIARETDB-001.

The CLAHE and ACL-CLAHE methods produced ab

the diabetic retinopathy condition, and it featured large nonuniform regions (i.e. vitreous area) and large detailed areas (i.e. vein and artery). However, this image suffered from a low contrast, which resulted in the obscurity of important details. Thus, this image was selected as a representative sample to assess the capabilities of all state-of-the-art technologies. For further evaluation, the analysis focused on two expanded regions (i.e., red and orange coloured boxed areas; Figures 24 and 25, respectively).

over-enhancement image, which led to over-contrast (Figures 23(b) and 23(h)). The vitreous area appeared darker, the vein and artery were nearly black, and the middle area became a black circle (Figure 23(b)). The presence of undesirable artifacts and noise in the output image caused difficulty in the observation of image details (Figures 24(b) and 25(b)). These outcomes hint that CLAHE and ACL-CLAHE are not the best options for enhancing the considered image type.

POSHE and Auto-CLAHE produced the poorest resultant images, which suffered from over-enhancement and unnatural appearance (Figure 23(c) and 23(g), respectively). The presence of undesirable artifacts and distortions in the resultant image caused difficulty in the observation of image details (red and orange coloured boxed regions in Figures 24(c) and 24(g)) and (25(c) and 25(g)), respectively. These observations imply that POSHE and Auto-CLAHE are the poorest options for enhancing the studied image type.

The BHM method produced an under-enhanced image with undesirable artifacts, which led to a low brightness (Figure 23(d)). Considering the red and orange coloured boxed region in Figures 24(d) and 25(d), respectively, observation of crucial image details presented difficulty. These findings imply that the BHM method is not the optimal option for enhancing the considered image type.

IAECHE and AEIHE produced dark and unpleasant resultant images (Figures 23(e) and 23(f), respectively). These methods failed to preserve crucial details and decreased image brightness. Consequently, the resulting images appeared slightly enhanced, represented by the red and orange coloured boxed regions in Figures 24(e) and 24(f) and 25(e) and 25(f). These observations hint that the IAECHE and AEIHE methods are not the optimal options for the enhancement of the investigated image type.

The proposed CLALHE method produced a resultant image with reduced noise or artifacts (Figure 23(i)) and exhibited a superior capability in preserving crucial image details (i.e. vitreous area, vein and artery; Figures 24(i) and 25(i)).

These observations confirm that the proposed CLALHE method, which effectively produced pleasant and natural images, is the optimum option for enhancing the considered image type.

The qualitative analysis above revealed the superior capability of the proposed CLALHE method in producing a high-quality image without artifacts, which makes it

Figures 20(i), 21(i) and 22(i), respectively. The qualitative analysis detailed above revealed the superior capability of the proposed CLALHE method in producing high-quality images with reduced artifacts, which makes it effective in detail preservation compared with other test methods. These observations confirm that the proposed CLALHE method is the optimum option for enhancing the investigated image type. It effectively produced pleasant and natural images, as evident in the background area (i.e. the sky), foreground areas (i.e., grass and trees) and large detailed areas (i.e. house, garden walls and car).

These findings are corroborated by the results of quantitative analysis. Table 6 tabulates the findings of quantitative analysis, where the CLALHE method produced the best PSNR, AMBE, SSI and RMSE values. Moreover, the proposed method yielded the second-best CII, which is close to 1 (0.999) and signifies its exceptional capability to preserve and enhance image details.

In conclusion, the combination of qualitative and quantitative analyses based on the input image (i.e. House 01) verified CLALHE's proficiency in producing pleasing and natural images and excellent CE and local detail preservation compared with other examination techniques.

For the last sample (i.e. DIARETDB1-001; Figure 23(a)), the original image was a medical image type that captures

efficient in detail preservation compared with the selected testing methods. These finding are corroborated by those of quan titative analysis. Table 7 displays the quantitative analysis results, which show that the CLALHE method attained the best PSNR, AMBE, SSI and RMSE values.

In conclusion, the combined qualitative and quantitative analyses based on the input image (i.e. DIARETDB1-001) verified CLALHE’s proficiency in producing pleasing and natural images and excellent CE and local detail preservation compared with other examination methods.

TABLE 8. Averages of the quantitative results for Faces 1999, Pasadena-Houses 2000 and DIARETDB1 (Note: the bolded values represent the best values, and the underlined values indicate the second-best values).

datasets (i.e. Faces 1999, Pasadena-Houses 2000 and DIARETDB1), an average quantitative analysis of all images enhanced by the CLALHE and other methods was conducted. Table 8 tabulates the averages of the quantitative analysis findings.

The proposed CLALHE method proved its superior capability in the production of pleasant and natural resultant images (Table 8).

The CLALHE method produced the best PSNR value for the Faces 1999 (31.822) and DIARETDB1 datasets (45.275), followed by Pasadena-Houses 2000 (31.020). The AEIHE technique attained the best PSNR of 32.534 for the Pasadena-Houses 2000, followed by the IAECHE method, which attained the second-best PSNR (29.361) for the Faces 1999 dataset. The Auto-CLAHE method achieved the second-best PSNR of 29.238 for the DIARETDB1 dataset.

The CLALHE method effectively conserved the mean brightness values, as evidenced by its best AMBE for all datasets, including the Faces 1999 (3.464), Pasadena Houses 2000 (2.197) and DIARETDB1 (1.224) datasets. Meanwhile, the second-best AMBE was evidenced by the AEIHE method for the Faces 1999 dataset (6.207), the AEIHE method for the Pasadena-Houses 2000 (6.358) and the POSHE method for the DIARETDB1 (2.948) dataset.

The CLALHE method effectively preserved image structure, as evidenced by the best SSI values (i.e. Faces 1999: 0.979, Pasadena-Houses 2000: 0.982; DIARETDB1: 0.975) it obtained. The second-best SSI values were evidenced by the BHM method for the Faces 1999 dataset (0.929), the BHM method for the Pasadena-Houses 2000 (0.931) and the IAECHE method for the DIARETDB1 (0.932).

TABLE 9. Average processing time analysis.

For further assessment of the outcomes for the selected

Moreover, the CLALHE method scored the best CII for the Pasadena-Houses 2000 dataset (1.0209), the best RMSE for the DIARETDB1 dataset (1.542) and the second-best entropy for the Pasadena-Houses 2000 dataset (7.511).

All these analyses proved the superior performance of the CLALHE method for local CE. This method provides resultant images with enhanced contrast, enriched information and conserved structures while minimising the effect of noise, artifacts and over enhancement.

IX. COMPUTATIONAL COMPLEXITY

This study analysed the computational complexity of the CLALHE algorithm through theoretical analysis (i.e. Big O notation). The computational complexity of the proposed method primarily hinges on the number of its operations and execution time. The CLALHE method involves multiple operations, including the following: (i) Reading and writing images requires taking the linear time proportional to the number of pixels, which results in the complexity of $O(M \cdot N)$, where M and N refer to the columns and rows of the image, respectively. (ii) Flattening the image takes linear time, which results in the complexity of $O(M \cdot N)$. (iii) Histogram calculation requires iteration over all pixels and their assignment to a histogram bin, which leads to a linear complexity of $O(M \cdot N)$. (iv) Finding peaks and valleys requires constant time, where the number of bins is constant (256), which results in a complexity of $O(b)$ where $b = 256$ (fixed bins). (v) CLAHE computation for peak values requires the peak and image sizes, where the CLAHE operation and PSNR, entropy and AMBE computation iterate over all image pixels for each peak, which results in a complexity of $O(P \cdot M \cdot N)$, where P = peaks number. (vi) Splitting the image into blocks, with each requiring the same operation, results in the complexity of $O(M \cdot N)$. (vii) The application of CLAHE for each block requires processing each subimage of size $M \cdot N$, which leads to a linear complexity of $O(M \cdot N)$. (viii) Merging of subimages involves their concatenation, which is linear in terms of total pixels and results in a linear complexity of $O(M \cdot N)$. Considering these factors, the overall computational complexity of the proposed method can be summarised as $O(M \cdot N)$. This linear complexity makes the algorithm suitable for image-processing tasks.

Furthermore, this study analysed the computational complexity of CLALHE algorithm through experimental simulation. Table 9 tabulates the average processing time for all comparisons. The analysis of processing times revealed the superior performance of the proposed CLALHE method compared with the other tested techniques. The table also provides the computational durations, which indicates that the CLALHE method achieved the shortest average processing times for image enhancement, specifically 0.58 s for Faces 1999, 1.92 s for Pasadena-Houses 2000 and 2.68 s for DIARETDB1. By contrast, other methods such as CLAHE, POSHE, BHM, Auto-CLAHE and ACL-CLAHE, required between 2 and 7 s of average processing time. In addition, the IAECH and AEIHE methods demonstrated longer

completion times (26 s to 83 s). These findings confirm that the CLALHE method outperforms other CE techniques in terms of processing time and affirm its suitability for real world applications.

X. CONCLUSION

This study presented a modified CLAHE method (CLALHE) designed to overcome the limitations of conventional techniques. CLALHE outperformed conventional methods by efficiently adapting parameter tuning, offering simplicity, and achieving low computational time. The CLALHE method introduced several improvements, including enhanced image contrast and local details, preservation of the overall image structure, and minimization of unwanted artifacts. Additionally, CLALHE adaptively identified optimal parameters (e.g., CBD and L_CL) through a multi-enhancement process and a new fitness function (CIQI) for selecting these parameters. It also subdivided the original image into subimages and enhanced each one using the optimal settings. We conducted qualitative and quantitative evaluations to assess the effectiveness of the local and adaptive features of CLALHE. Metrics, such as PSNR, entropy, AMBE, SSI, CII and RMSE, were used to validate the results. The CLALHE method produced enhanced images with improved local details, preserved structure and minimizing noise. In addition, the CLALHE method remarkably enhances low-contrast images. It enables the production of images with preserved details and reduced artifacts. These characteristics make CLALHE a valuable tool for various image-processing applications, specifically real-time applications. This work also highlighted CLALHE as an optimum solution for the enhancement of image contrast under nonuniform illumination and varying levels of detail. Moreover, the proposed method outperforms other CLAHE-based methods in qualitative and quantitative evaluations, demonstrating the effectiveness of our proposed method and improvement. Future research should explore optimisations, real-time processing and integration with other image processing techniques to further enhance the applicability and efficiency of the CLALHE method.

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